

MY457/MY557: Causal Inference for Observational and Experimental Studies

Week 8: Instrumental Variables 2

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Course Outline

- **Week 1:** Causal frameworks
- **Week 2:** Randomization
- **Week 3:** Selection on observables I
- **Week 4:** Selection on observables II
- **Week 5:** Selection on observables III
- *Week 6: Reading week*
- **Week 7:** Instrumental variables I
- **Week 8:** Instrumental variables II
- **Week 9:** Regression discontinuity
- **Week 10:** Difference-in-differences I
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- 1 Examiner Instruments
- 2 Shift-Share Instruments
- 3 Recentered Instruments

Motivating Example: Examiner Instruments

Does short-term incarceration affect downstream political participation?

Any naïve study of this would likely be **confounded**:

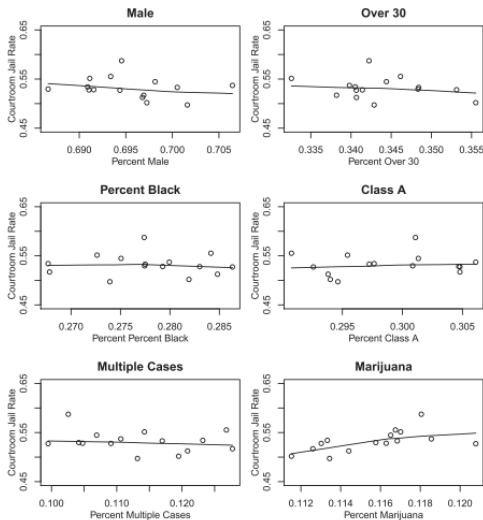
- Generally, those who spend time in jail are dissimilar to those who do not.
- Could study only those prosecuted, using **legal guilt as instrument**. But juries (determining legal guilt) are likely systematically biased.
- Could study only those found guilty, studying **jail-time directly**. But judges (sentencing people to jail-time) are likely systematically biased too.

White (2019) studies Harris County (Texas) where:

- First time defendants are **randomly** assigned to courtrooms
- Courtrooms differ in the **propensity for defendants to serve jail-time**
- This is called an **examiner ('judge') instrument**

Motivating Example: Examiner Instruments

FIGURE 1. Scatterplots of Pre-treatment Case Characteristics Against Courtroom Incarceration Rates.



Setup: Examiner Instruments

Let's generalise this research setting:

- N units (typically individuals), indexed by $i = 1, \dots, N$
- K examiners **control treatment** status $D_i \in \{0, 1\}$, which may affect outcome of interest Y_i
- Each examiner has L_k , a **fixed latent propensity for assigning treatment**

Each i is **assigned an examiner in a known way** s.t. they receive status $Q_i \in \{0, \dots, K - 1\}$, from which we can write $Z_{ki} = \mathbf{1}[Q_i = k]$

Given this we can define **potential outcomes**:

- D_{qi} : p.o. of treatment for i under examiner status q
- Y_{qi} : p.o. of outcome for i under examiner status q
- Y_{dqi} : p.o. of outcome for i under treatment d and examiner status q

We now have a **high-dimensional** instrument(s) for one treatment, what is called **overidentification**.

May face a **many weak instruments problem** $\rightsquigarrow$ significant **bias**. (More in a moment!)

Illustration with Potential Outcomes

For now, consider a toy example with three units and three examiners:

i	Q_i	L_q	Z_i^0	Z_i^1	Z_i^2	D_{qi}	Y_{dqi}
1	0	0.4	1	0	0	0	1
1	1	0.5	0	1	0	0	1
1	2	0.8	0	0	1	0	0
2	0	0.4	1	0	0	0	0
2	1	0.5	0	1	0	1	0
2	2	0.8	0	0	1	1	0
3	0	0.4	1	0	0	1	1
3	1	0.5	0	1	0	0	1
3	2	0.8	0	0	1	1	1

Note: Here we represent the **latent propensity for treatment** as L_q rather than L_k – it is the latent propensity that i is assigned **to** given Q_i . This is equivalent to writing $L_q = \sum_{k=0}^{K-1} L_k \mathbf{1}[Q_i = k]$

Identification Assumptions: Examiner IV

What do our IV assumptions look like in the examiner design?

1. SUTVA
2. **Relevance**: Does every examiner assign D differently?
3. **Ignorability**: May be given, but often only holds conditionally.
4. **Exclusion**: What else can examiners control?
 - ↪ Can be weakened to an **on average exclusion restriction**, s.t. any direct examiner effects are independent of their first-stage effects.
5. **Monotonicity**: Examiner behaviour must be ordered.
 - ↪ If examiner 0 more lenient on average, must be weakly more lenient for **every i** !
 - ↪ Example violation: examiner 0's leniency varies based on unit-characteristics, e.g. racial or gender biases.
 - ↪ Can be weakened to **average monotonicity**: units may violate monotonicity with some examiners, so long as they **comply on average**.

Average Assumptions for Examiner IV

Consider again our toy example:

i	Q_i	L_q	Z_i^0	Z_i^1	Z_i^2	D_{qi}	Y_{dqi}
1	0	0.4	1	0	0	0	1
1	1	0.5	0	1	0	0	1
1	2	0.8	0	0	1	0	0
2	0	0.4	1	0	0	0	0
2	1	0.5	0	1	0	1	0
2	2	0.8	0	0	1	1	0
3	0	0.4	1	0	0	1	1
3	1	0.5	0	1	0	0	1
3	2	0.8	0	0	1	1	1

Here we have a strict monotonicity violation (unit 3, examiners 0 and 1), but we still have **average monotonicity**.

We also have an exclusion violation (unit 1, examiner 1), but we **may** still have **average exclusion**.

See Frandsen et al (2023) for more, including tests.

Latent Variable Estimation

Identifying variation in treatment comes from **examiners' varying L_k** , and (as-if, possibly conditional) random assignment of units to examiners.

Intuitively, we might **estimate this quantity directly**, e.g.:

$$\hat{L}_{ik} = \frac{\sum_{j \neq i} \mathbf{1}[Q_j = Q_i] D_j}{\sum_{j \neq i} \mathbf{1}[Q_j = Q_i]}$$

Read: Average realised D assigned by k , ignoring the realised D for unit i .

Note: This means $\hat{L}_{ik}$ will **vary both by k and i** , though the latent variable L_k is only k -varying. This is called a **leave-one-out** or **jackknife** procedure, and solves the **many weak instruments** bias problem. (Why?)

We can thus **instrument for D_i using $\hat{L}_{ik}$** with 2SLS. But note:

1. Under what constraints is examiner assignment, and thus $\hat{L}_{ik}$ random?
2. Variability in our estimates of $\hat{L}_{ik}$? Think $\frac{N}{K-1}$.
3. Uncertainty in estimates of $\hat{L}_{ik}$?

Examiner Fixed-Effects

Observe that L_k is an examiner-specific property, and can be subsumed under an **invariant examiner-specific 'fixed-effect'**.

Can thus use $Z^1, \dots, Z^k$ as multiple dummy variable instruments in first stage rather than $\hat{L}_{ik}$.

Include conditioning set directly as additional controls.

- Any conditioning set must be included in **both** stages
- In the latent estimation approach, conditioning is dealt with **when estimating the latent property**, resolving issue 1 on the previous slide. Additional controls can be included in 2SLS specification.

Estimate using **Jackknife Instrumental Variables Estimation** (JIVE) (Angrist et al, 1999) rather than 2SLS.

With no covariates, JIVE and latent estimation are $\approx$ equivalent, both use **leave-one-out** to avoid bias.

If you have **covariates**, use **UJIVE** (Kolesar, 2013) instead.

Latent Estimation vs. Fixed Effects?

So, should we directly estimate L_k , or use fixed effects?

Fixed-effects with JIVE or UJIVE is **probably preferable**:

- Transparency: Directly maps the 'experiment' to your estimator.
- Covariates: More straightforward to adjust directly in regression.
- Standard errors: We should account for uncertainty in $\hat{L}_{ik}$, which adds an additional inferential challenge.
- Note: If you want to 'show' variation in the instrument, you can still do this using first stage coefficients on your $Z^1, \dots, Z^k$ dummies.

These days, as marginal computational and coding costs approach zero, there is no good argument against doing and reporting both. Your results should not hinge on **(approximately arbitrary) estimation choices**.

Illustration: Realised Data

Realized data might look something like this:

i	Y_i	D_i	Q_i	$\hat{L}_{ik}$	Z_i^0	Z_i^1	$\dots$	Z_i^{K-1}
1	1	1	1	$\hat{I}_{11}$	1	0	$\dots$	0
2	0	1	2	$\hat{I}_{22}$	0	1	$\dots$	0
3	1	0	1	$\hat{I}_{31}$	1	0	$\dots$	0
4	1	1	K	$\hat{I}_{4K}$	0	0	$\dots$	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
N	0	1	1	$\hat{I}_{N1}$	1	0	$\dots$	0

Note: Overidentification here is given by the fact that **all K_i^1 through K_i^K variables** (or dimensions) are jointly determining a unit's D_i status. That is, there is a **many-to-one** instrument-to-endogenous-regressor mapping.

Back to the Motivating Example: Estimation and Results

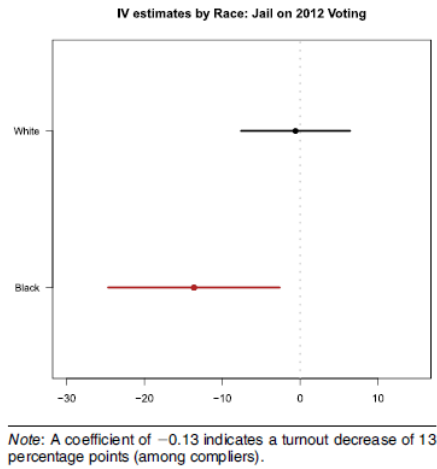
White (2019) uses the first approach (latent estimation):

1. Estimate courtroom-year **average incarceration rate**
(in appendix shows courtroom-year FE approach)
2. 2SLS model with **courtroom-year rate as instrument** and year-dummies
3. Subset by **race** of defendant for heterogeneous effects

TABLE 3. Jail Sentences on 2012 Voting		
	<i>Dependent variable</i>	
	Jail (1)	Voted 2012 (2)
Court jail average (Yr)	1.000* (0.051)	
Jail		−0.045 (0.034)
Constant	−0.0001 (0.029)	0.142* (0.019)
Year dummies	Yes	Yes
Observations	113,367	113,367
Adjusted R^2	0.004	0.017
F statistic	97.948* (df = 5; 113,361)	
Note: * $p < 0.05$.		

Back to the Motivating Example: Heterogeneous Effects

FIGURE 2. Jail's Effect on Voter Turnout (2SLS Estimates), by Race of Defendant.



A Motivating Example: Shift-Share Instruments

How does in-migration of **new** minority groups affect social position of **pre-existing** minority groups?

Fouka, Mazumder, & Tabellini (2022) study the context of the **First Great Migration** in the USA at the MSA-level:

- 1915 - 1930, 1.5m Black Americans left Southern for Northern states.
- Did this affect white Americans' views of European immigrants?

Much like Ananat (2011), many potential **confounders** exist. Black Americans may move to...

- ... more liberal MSAs
- ... wealthier MSAs
- ... etc.

Authors use a shift-share instrumental variables (SSIV) approach.

Setup: The Endogenous Model

Consider a simple model for $l = 1, \dots, L$ units (labour markets, areas, MSAs etc.):

$$y_l = \alpha + \beta x_l + \varepsilon_l$$

where β is the effect of x_l (our endogenous regressor) on y_l
and we assume it is confounded ($\mathbb{E}[x_l \varepsilon_l] \neq 0$).

Note: We are using the **classical IV** approach here. Both ignorability and exclusion fall under a single '**valid instrument**' assumption. For instrument z_l , satisfied if $\mathbb{E}[z_l \varepsilon_l] = 0$, equivalently $\text{Cov}(z_l, \varepsilon_l) = 0$ or $\mathbb{E}[\varepsilon_l | z_l] = 0$.

Now imagine we had a **second dimension of observation** for our **units**, across $k = 1, \dots, K$ which are **types** (industry, population groups, migrants from specific places, etc.).

Each unit l can be observed for all or multiple k , so we have y_{lk} and x_{lk} .

Setup: Shift-Share IV

Given both y_{lk} and x_{lk} , we have two sources of potential variation:

1. **Shares**: A unit- and type-varying (lk -varying) baseline variable, z_{lk}
2. **Shift(s)**: A type-varying (k -varying) change variable, g_k

Interacting the above gives us a **shift-share** variable: $S_l = \sum_k z_{lk} \times g_k$

Read: The mapping of **exposure** (share) to a **shock** (shift), capturing unit-varying shock-exposures.

Intuition: Use this interaction to instrument for x_l .

Note: Shift-share instruments are sometimes called **Bartik instruments**, after Bartik (1991).

Back to the Motivating Example: Shift-Share Instrument

Fouka, Mazumder, & Tabellini (2022) construct their instrument as:

1. **Shares**: z_{lk} , the pre-shift share of Black American migrants born in state k living in MSA l
2. **Shift(s)**: g_k , the share of Black Americans born in state k who left
3. **Shift-share**: $S_l = \sum_k z_{lk} \times g_k$, the predicted post-shift sum of Black American migrants in MSA l

Read: Their instrument captures the predicted level of Black American in-migration in an MSA, based on **some randomness in the timing of out-migration from different states**.

Use this to instrument for realised Black American in-migration by decade.

Back to the Motivating Example: First Stage

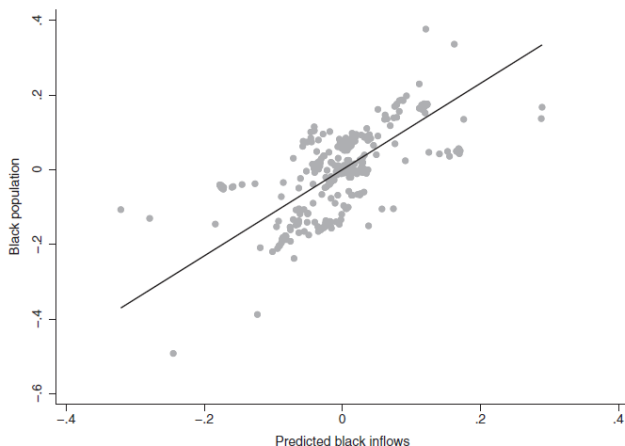


FIGURE 3
First stage

Notes: The figure shows the relationship between actual and predicted Black population for the years 1910–30. Each point represents the residual change in an MSA's actual (y-axis) and predicted (x-axis) number of Black migrants after partialling out total MSA population and MSA and region by decade fixed effects.

Back to the Motivating Example: Results

TABLE 2
Effects of Black inflows on intermarriage and naturalization rates

	(1) OLS	(2) OLS	(3) 2SLS	(4) 2SLS	(5) 2SLS	(6) 2SLS
Panel A: Married w/ native (1910 mean: 0.071)						
Black population	0.007 (0.006)	0.005 (0.004)	0.012** (0.005)	0.013*** (0.005)	0.012** (0.005)	0.019 (0.017)
Observations	9,323,126	9,323,126	9,323,126	9,323,109	9,323,109	88,892
F-stat			23.33	23.71	23.83	32.75
Panel B: Naturalized (1910 mean: 0.491)						
Black population	0.056** (0.025)	0.030* (0.016)	0.033** (0.015)	0.038*** (0.013)	0.033** (0.014)	0.144*** (0.031)
Observations	15,267,846	15,267,846	15,267,846	15,267,844	15,267,844	80,866
F-stat			24.23	24.38	24.51	32.74
Individual controls	X	X	X	X	X	
Region × Decade		X	X	X	X	
MSA × Origin region				X	X	
Origin region × Decade					X	
Linked sample						X

Notes: The table presents results for immigrant men living in the 108 non-southern MSAs for which the instrument could be constructed in census years 1910, 1920, and 1930. In Panel A, the sample is restricted to married men. *Married w/ native* is a dummy equal to 1 if the individual is married to a native-born spouse of native-born parentage. Columns 1–5 present results obtained from the repeated cross-sections, while column 6 shows results from the linked panel of men who always remained in the same MSA in the three census years. Columns 1–2 (resp. 3–6) present OLS (resp. 2SLS results). Individual controls include fixed effects for age, years in the U.S. and origin region. All regressions control for MSA and year fixed effects and for total MSA population. Regressions in column 6 include individual fixed effects. In column 6 of Panel A, the sample is restricted to men who were not married in the previous decade (Panel A) or who were not naturalized in the previous decade (Panel B). F-stat refers to the KP F-stat for weak instruments. Robust standard errors, clustered at the MSA level, in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Identification Assumptions: Shares Perspective

There are two broad perspectives on identification in shift-share settings.

First relies on **exposure mapping** via **shares** (Goldsmith-Pinkham et al (2020)):

1. Relevance of the share:
 - ↪ For all k , z_{lk} must predict x_l
2. Strict exogeneity of the share to the error term, $\mathbb{E}[z_{lk}\varepsilon_l] = 0$
 - ↪ This could be a conditional statement, or indexed over time
 - ↪ Intuition: exogeneity is w.r.t. **changes** in, not the level of, y

With these two assumptions (plus SUTVA), **two stage least squares** with the shift-share instrument is **consistent**.

Identification Assumptions: Shifts Perspective

Second approach relies on **shifts** ('**shocks**' in Borusyak et al (2022)):

1. Exogeneity of the shocks to confounding in the shares, $\mathbb{E}[\mathbf{g}_k | \bar{\mathbf{e}}\mathbf{z}] = \mu$ for all k
 $\rightsquigarrow$ Every shock $\mathbf{g}_k$ has the same mean μ , regardless of observables ($\bar{\mathbf{e}}$) or shares ($\mathbf{z}$). Essentially, shocks are as-if randomly assigned.
2. Many uncorrelated shocks: $\mathbb{E}[\sum_k \mathbf{z}_k^2] \rightarrow \mathbf{0}$ and $\text{Cov}(\mathbf{g}_k, \mathbf{g}_{k'}) = \mathbf{0}$ for all (k, k') with $k \neq k'$.
 $\rightsquigarrow$ Intuition: with very many exogenous shocks, residual confounding bias from shares averages out.

With these two assumptions plus relevance and SUTVA, **two stage least squares** is **consistent**.

Shifts or Shares?

Consider a few settings:

1. Cross-sectional (no time-variation), two types ($K = 2$):
 $\rightsquigarrow$ Identification tends to rely on exogeneity of **shares**
2. Multiple (T) time periods, two types:
 $\rightsquigarrow$ Identification tends to rely on **shares**
3. Many ($K > 2$) types, cross-sectional or multiple T :
 $\rightsquigarrow$ For **very many K** , rely on exogeneity of **shocks**

See Goldsmith-Pinkham et al (2020) (focused on shares) and Borusyak et al. (2022) (focused on shifts) for various falsification tests.

In general, be explicit (in reasoning and language) about **what variation you think facilitates identification**.

A Motivating Example: Recentered Instruments

Does distributing government funds affect election outcomes?

This is hard to study because... ok by now you're tired of me saying: **confounding**. But again, it's a problem. Targeted areas might be...

- ... wealthier (or not)
- ... politically aligned
- ... etc.

Gulotty & Strezhnev (2024) study the US Department of Agriculture's (USDA) Market Facilitation Program (MFP) and the 2020 US election.

In 2019, the MFP allocated \$14bn to agricultural producers using a formula based on:

1. Measures of agricultural production ← **endogenous**
2. Product-specific trade damage estimates ← **plausibly random?**

Setup: Recentered IV

For $i = 1, \dots, N$, assume a simple model:

$$y_i = \beta x_i + \varepsilon_i$$

Consider $\mathbf{z}_i = f_i(\mathbf{g}, \mathbf{w})$ as a candidate instrument for x_i , where:

- $\mathbf{g}$ is a vector of shocks
- $\mathbf{w}$ represents possibly endogenous variables (e.g. shares)
- $f_i(\cdot)$ maps the shocks and endogenous variables to $\mathbf{z}_i$
- Allow $\mathbb{E}[\mathbf{z}_i \varepsilon_i] \neq \mathbf{0}$ (bad!)

Note: $\mathbf{z}_i$ need not be an instrument, could be a treatment

Intuition: While total variation in $\mathbf{z}_i$ may not be useful (see the third bullet point above), some variation in $\mathbf{z}_i$ may be (as good as) **random** while some is **systematic (endogenous)**.

Similar flavour to shift-share IV!

Identification Assumptions and Result: Recentered IV

Assume:

1. Exogenous shocks s.t. $\mathbf{g} \perp\!\!\!\perp \varepsilon \mid \mathbf{w}$
2. Distribution $\mathbf{G}(\mathbf{g}|\mathbf{w})$ is known $\leftarrow$ this is **important!**

We can write down the expected value (average) of all potential shocks as:

$$\mu_i = \mathbb{E}[f_i(\mathbf{g}, \mathbf{w})]$$

Under **assumption 1**, can show that:

$$\mathbb{E} \left[\frac{1}{N} \sum_i \mathbf{z}_i \varepsilon_i \right] = \mathbb{E} \left[\frac{1}{N} \sum_i \mu_i \varepsilon_i \right]$$

This suggests a 'recentering' result:

$$\mathbb{E} \left[\frac{1}{N} \sum_i \tilde{\mathbf{z}}_i \varepsilon_i \right] = \mathbb{E} \left[\frac{1}{N} \sum_i \mathbf{z}_i \varepsilon_i \right] - \mathbb{E} \left[\frac{1}{N} \sum_i \mu_i \varepsilon_i \right] = 0$$

Under **assumption 1** β is identified by $\tilde{\mathbf{z}}_i$ (assuming **relevance** of $\tilde{\mathbf{z}}_i$), and under **assumption 2** it can be **unbiasedly estimated** with data.

Estimation: Recentered IV

Borusyak & Hull (2023) propose a simple method for purging the **systematic** component:

1. Precisely state the formula that generates $\mathbf{z}_i$
2. Using knowledge of **random** component, define counterfactual **shocks** (by assumption 2)
3. Calculate $\hat{\mu}_i = \text{average } \mathbf{z}_i$ across simulations of counterfactual shocks
 $\rightsquigarrow$ this represents the **systematic component**
 $\rightsquigarrow$ can think of it as akin to a propensity score
4. Recenter $\mathbf{z}_i$ to create $\tilde{\mathbf{z}}_i = \mathbf{z}_i - \hat{\mu}_i$
5. Instrument with $\tilde{\mathbf{z}}_i$ (or use $\mathbf{z}_i$ directly, controlling for $\hat{\mu}_i$)
6. Use the counterfactual shocks for randomization inference to generate p-values etc.

Back to the Motivating Example: Treatment via Formula

Recall Gulotty & Strezhnev (2024) wish to study the effect of MFP agricultural transfers to counties.

The authors are able to **reconstruct** the USDA **formula for MFP** as:

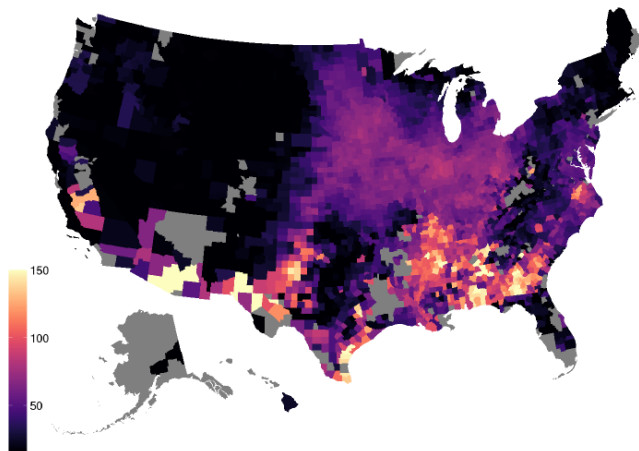
$$(\text{County Rate})_{\$/\text{acre}} = \frac{\sum_{c=1}^C (\text{County Acres})_{\text{acre}}^c \times (\text{County Yield})_{\text{unit}/\text{acre}}^c \times (\text{Crop Rate})_{\$/\text{unit}}^c}{\sum_{c=1}^C (\text{County Acres})_{\text{acre}}^c}$$

Where the critical component $(\text{Crop Rate})_{\$/\text{unit}}$ is given by:

$$(\text{Crop Rate})_{\$/\text{unit}} = \frac{(\text{Value of Exports})_{\$} \times (\% \text{ Reduction in exports})}{(\text{Annual Production})_{\text{unit}}}$$

The numerator comes from product-specific trade damage estimates which rely on historical export volatility, **plausibly unconnected** to domestic politics.

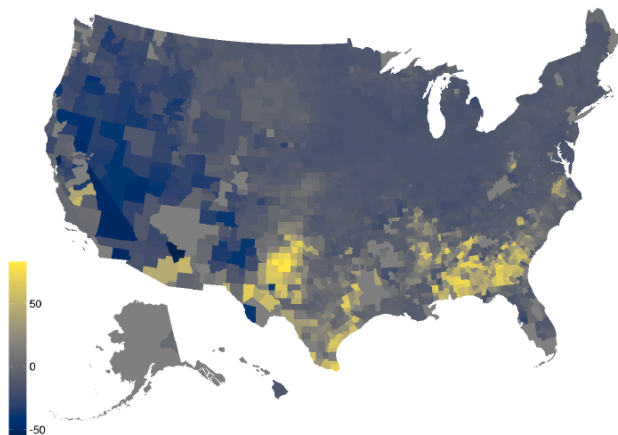
Back to the Motivating Example: Treatment via Formula



Notes: 2,892 counties. In the few cases where separate rates were calculated for parts of the same county, we combined the USDA's total acreage and payment calculations for each component to construct an "average" county rate.

Figure 1: Map of actual USDA 2019 non-specialty crop MFP payment rates (dollars per acre)

Back to the Motivating Example: Recentered Instrument

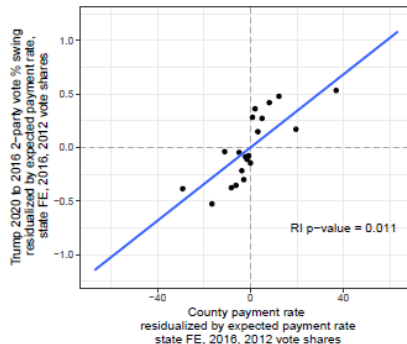


Notes: 2,877 counties with non-zero eligible acres based on FSA crop acreage data. “Expected” county rates calculated via 3000 random permutations of the per-acre commodity excess rate shocks.

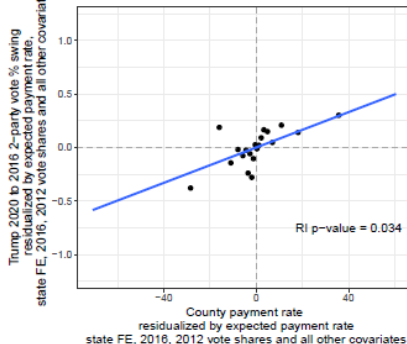
Figure 4: Map of recentered|county rates (dollars per acre)

Back to the Motivating Example: Results

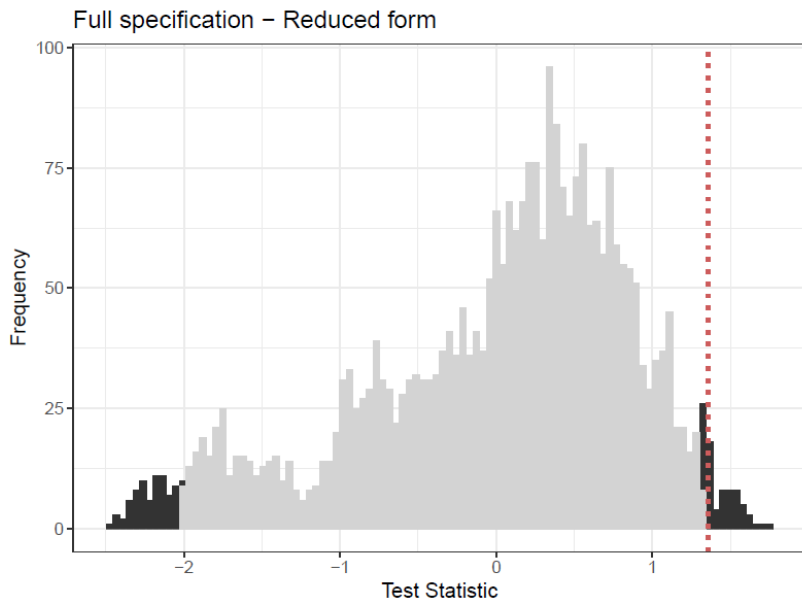
Limited specification



Full specification



Back to the Motivating Example: Randomization Inference



Thanks to Kirill Borusyak, Paul Goldsmith-Pinkham, and Peter Hull, whose publicly available materials I leaned heavily on for this week.